Assessment Menu: Teacher Guide

The Right Triangle Bistro · design notes, grading and answer key

Why students cannot just pick the easy ones

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

Skill group	What it covers	Dishes
PYTHAG	Pythagorean theorem and its converse	A1, A2, E1, E2, S1
SPECIAL	Special right triangles (45°-45°-90° and 30°-60°-90°)	A3, A4, E4, S2
TRIG	Sine, cosine and tangent	A5, E5, E6, S3
APPLY	Applications and angles of elevation or depression	A6, E3, S4

If a student only does…	Maximum
Only Appetizers (4 × 5)	20
Appetizers + Entrées (4 × 5 + 4 × 10)	60
Full meal (adds 2 Chef's Specials)	100

Grading and options

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

Student	Appetizers (of 20)	Entrées (of 40)	Specials (of 40)	Skills OK?	Total

Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
A1 PYTHAG	A right triangle has legs 9 and 12. Find the length of the hypotenuse.	**Answer:** 15 $c^2 = 9^2 + 12^2 = 81 + 144 = 225$ $c = \sqrt{225} = 15$	3 pts: Sets up $a^2 + b^2 = c^2$ with the legs 2 pts: Hypotenuse 15
A2 PYTHAG	Could 7, 24, and 25 be the side lengths of a right triangle? Show your work.	**Answer:** Yes: $7^2 + 24^2 = 625 = 25^2$ The longest side must be the hypotenuse: test $7^2 + 24^2 = 49 + 576 = 625$. $25^2 = 625$, so the converse of the Pythagorean theorem says it is a right triangle.	3 pts: Computes $49 + 576 = 625$ and $25^2 = 625$ 2 pts: Concludes yes (right triangle) by the converse
A3 SPECIAL	A 45°-45°-90° triangle has legs of length 5. Find the hypotenuse in simplest radical form.	**Answer:** $5\sqrt{2}$ In a 45°-45°-90° triangle, hypotenuse = leg × $\sqrt{2}$. Hypotenuse = $5\sqrt{2}$.	2 pts: Uses hypotenuse = leg × $\sqrt{2}$ (or the Pythagorean theorem) 3 pts: Correct answer $5\sqrt{2}$
A4 SPECIAL	A 30°-60°-90° triangle has a shorter leg of 4. Find the hypotenuse and the longer leg.	**Answer:** Hypotenuse 8, longer leg $4\sqrt{3}$ Hypotenuse = 2 × shorter leg = 8. Longer leg = shorter leg × $\sqrt{3} = 4\sqrt{3}$.	2 pts: Hypotenuse 8 3 pts: Longer leg $4\sqrt{3}$
A5 TRIG	In $\triangle ABC$, $\angle C = 90°$, AB = 10, BC = 6, and AC = 8. Write sin A, cos A, and tan A as fractions in lowest terms.	**Answer:** $\sin A = \frac{3}{5}$, $\cos A = \frac{4}{5}$, $\tan A = \frac{3}{4}$ $\sin A = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{6}{10} = \frac{3}{5}$ $\cos A = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{8}{10} = \frac{4}{5}$ $\tan A = \frac{\text{opposite}}{\text{adjacent}} = \frac{6}{8} = \frac{3}{4}$	2 pts: $\sin A = \frac{3}{5}$ 1 pts: $\cos A = \frac{4}{5}$ 2 pts: $\tan A = \frac{3}{4}$ (opposite and adjacent are relative to $\angle A$)
A6 APPLY	A 13 ft ladder leans against a wall and reaches 12 ft up the wall. How far is the base of the ladder from the wall?	**Answer:** 5 ft The ladder is the hypotenuse: $d^2 + 12^2 = 13^2$ $d^2 = 169 - 144 = 25$, so d = 5 ft.	3 pts: Sets up $d^2 + 12^2 = 13^2$ (ladder is the hypotenuse) 2 pts: d = 5 ft

Answer Key: Entrées (10 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
E1 PYTHAG	A rectangle has a diagonal of 17 and a width of 8. Find its length and its perimeter.	**Answer:** Length 15; perimeter 46 The diagonal is the hypotenuse of a right triangle with legs 8 and L. $L^2 = 17^2 - 8^2 = 289 - 64 = 225$, so $L = 15$. Perimeter $= 2(15 + 8) = 46$.	4 pts: Treats the diagonal as the hypotenuse: $L^2 + 8^2 = 17^2$ 2 pts: Length 15 4 pts: Perimeter 46
E2 PYTHAG	Which of these could be the sides of a right triangle? Show the test for each. (a) 10, 24, 26 (b) 9, 40, 42 (c) 12, 35, 37	**Answer:** (a) yes (b) no (c) yes Test $a^2 + b^2$ against the longest side squared. (a) $100 + 576 = 676 = 26^2$ ✓ right (b) $81 + 1600 = 1681$, but $42^2 = 1764$ ✗ not right (c) $144 + 1225 = 1369 = 37^2$ ✓ right	2 pts: Uses the longest side as c in each test 3 pts: (a) correct test and conclusion 2 pts: (b) $1681 \neq 1764$, not right 3 pts: (c) correct test and conclusion
E3 APPLY	A wheelchair ramp rises 1.5 ft over a horizontal run of 18 ft. Find the angle of the ramp with the ground and the length of the ramp surface, each to the nearest tenth.	**Answer:** Angle ≈ 4.8°; length ≈ 18.1 ft $\tan\theta = \frac{1.5}{18} = \frac{1}{12}$, so $\theta = \tan^{-1}(\frac{1}{12}) \approx 4.8°$. Length $= \sqrt{(1.5^2 + 18^2)} = \sqrt{326.25} \approx 18.1$ ft.	3 pts: $\tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{1.5}{18}$ 2 pts: $\theta \approx 4.8°$ using inverse tangent 3 pts: Pythagorean theorem set up 2 pts: Length ≈ 18.1 ft
E4 SPECIAL	An equilateral triangle has side length 10. Draw an altitude and use a special right triangle to find the altitude and the area of the triangle, in simplest radical form.	**Answer:** Altitude $5\sqrt{3}$; area $25\sqrt{3}$ The altitude splits the triangle into two 30°-60°-90° triangles with hypotenuse 10 and shorter leg 5. Altitude $= 5\sqrt{3}$ (shorter leg × √3). Area $= \frac{1}{2} \times 10 \times 5\sqrt{3} = 25\sqrt{3}$.	4 pts: Shorter leg 5 and 30°-60°-90° reasoning 3 pts: Altitude $5\sqrt{3}$ 3 pts: Area $25\sqrt{3}$
E5 TRIG	In △PQR, ∠Q = 90°, ∠P = 35°, and PR = 20. Find QR and PQ to the nearest tenth. (sin 35° ≈ 0.5736, cos 35° ≈ 0.8192)	**Answer:** QR ≈ 11.5; PQ ≈ 16.4 QR is opposite ∠P: $\sin 35° = \frac{QR}{20}$, so $QR = 20(0.5736) \approx 11.5$. PQ is adjacent to ∠P: $\cos 35° = \frac{PQ}{20}$, so $PQ = 20(0.8192) \approx 16.4$.	2 pts: Chooses sine for QR 3 pts: QR ≈ 11.5 2 pts: Chooses cosine for PQ 3 pts: PQ ≈ 16.4
E6 TRIG	In right △ABC (∠C = 90°) the side opposite ∠A is 5 and the hypotenuse is 13. Kai writes $\tan A = \frac{5}{13}$, so $A = \tan^{-1}(\frac{5}{13}) \approx 21°$. Find his mistake and give the correct angle to the nearest degree.	**Answer:** Tangent uses the adjacent leg, not the hypotenuse. $\sin A = \frac{5}{13}$, so $A \approx 23°$. Tangent is $\frac{\text{opposite}}{\text{adjacent}}$; 13 is the hypotenuse, so Kai used the wrong ratio. Opposite and hypotenuse give sine: $\sin A = \frac{5}{13}$ $A = \sin^{-1}(\frac{5}{13}) \approx 22.6° \approx 23°$.	4 pts: Identifies the mistake: tangent needs the adjacent leg, but 13 is the hypotenuse 3 pts: Uses $\sin A = \frac{5}{13}$ 3 pts: $A \approx 23°$

Answer Key: Chef’s Specials (20 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
S1 PYTHAG	In △ABC, D is on side BC, AD ⊥ BC, AB = 13, AC = 20, and AD = 12. Find BD, DC, BC, and the perimeter of △ABC. Then use the converse of the Pythagorean theorem to decide whether △ABC is a right triangle.	**Answer:** BD = 5, DC = 16, BC = 21, perimeter 54; not a right triangle BD = $\sqrt{(13^2 - 12^2)} = \sqrt{25} = 5$. DC = $\sqrt{(20^2 - 12^2)} = \sqrt{256} = 16$. BC = 5 + 16 = 21. Perimeter = 13 + 20 + 21 = 54. Converse test with the longest side 21: $13^2 + 20^2 = 169 + 400 = 569$, but $21^2 = 441$. Not equal, so △ABC is **not** a right triangle.	8 pts: BD = 5 (4) and DC = 16 (4) using right triangles ABD and ADC 4 pts: BC = 21 and perimeter 54 5 pts: Converse test with the longest side as c (169 + 400 vs 441) 3 pts: Correct conclusion: not a right triangle
S2 SPECIAL	Rectangle ABCD has diagonal AC = 20 and ∠CAB = 30°. (a) Explain why △ABC is a 30°-60°-90° triangle. (b) Find AB and BC in simplest radical form. (c) Find the area and perimeter of the rectangle.	**Answer:** (b) BC = 10, AB = 10√3 (c) area 100√3, perimeter 20 + 20√3 (a) ∠B = 90° (rectangle) and ∠CAB = 30°, so ∠ACB = 60°. (b) BC is opposite 30°, so BC = $\frac{1}{2} \times 20 = 10$. AB = 10√3. (c) Area = 10 × 10√3 = 100√3. Perimeter = 2(10 + 10√3) = 20 + 20√3.	4 pts: (a) Right angle at B and angles 30°, 60°, 90° 8 pts: (b) BC = 10 (4); AB = 10√3 (4) 8 pts: (c) Area 100√3 (4); perimeter 20 + 20√3 (4)
S3 TRIG	In right △ABC, ∠C = 90°, $\sin A = \frac{7}{25}$, and AB = 50. (a) Find BC. (b) Find AC. (c) Find cos A and tan A. (d) Check your sides with the Pythagorean theorem.	**Answer:** (a) BC = 14 (b) AC = 48 (c) $\cos A = \frac{24}{25}$, $\tan A = \frac{7}{24}$ (a) $\sin A = \frac{BC}{AB} : \frac{BC}{50} = \frac{7}{25}$, so BC = 14. (b) $AC^2 = 50^2 - 14^2 = 2500 - 196 = 2304$, so AC = 48. (c) $\cos A = \frac{48}{50} = \frac{24}{25}$; $\tan A = \frac{14}{48} = \frac{7}{24}$. (d) $14^2 + 48^2 = 196 + 2304 = 2500 = 50^2$ ✓	4 pts: (a) BC = 14 5 pts: (b) AC = 48 with work 6 pts: (c) $\cos A = \frac{24}{25}$ (3); $\tan A = \frac{7}{24}$ (3) 5 pts: (d) 196 + 2304 = 2500 shown
S4 APPLY	A boat is 120 m from the base of a sea cliff. The angle of elevation to the top of the cliff is 25°, and to the top of a lighthouse at the cliff’s edge it is 32°. (tan 25° ≈ 0.4663, tan 32° ≈ 0.6249) (a) Find the cliff height and the lighthouse top height above sea level, to the nearest tenth. (b) Find the lighthouse height. (c) From the lighthouse top, what is the angle of depression to the boat, and why?	**Answer:** (a) 56.0 m and 75.0 m (b) 19.0 m (c) 32°, alternate interior angles (a) Cliff: $\tan 25° = \frac{h}{120}$, so h = 120(0.4663) ≈ 56.0 m. Top of lighthouse: 120(0.6249) ≈ 75.0 m above sea level. (b) Lighthouse = 75.0 − 56.0 = 19.0 m. (c) The angle of depression is 32°: the horizontal line from the top is parallel to the sea, so it equals the angle of elevation (alternate interior angles).	5 pts: (a) Cliff: $\tan 25° = \frac{h}{120}$ (3); 56.0 m (2) 5 pts: (a) Top: tan 32° equation (3); 75.0 m (2) 4 pts: (b) Subtracts to get 19.0 m 6 pts: (c) 32° (2); explains parallel lines and alternate interior angles (4)